Systems working together

Practice Paper 2 | 30 marks | Suggested time: 35 minutes

Teacher/student marking guidance. Award credit for equivalent scientifically correct wording. Marks within each part are capped at the stated total.

Question 1 Circulation — 8 marks

a) For arteries, veins and capillaries, state ONE distinctive feature or function of each. **[3]**

- Artery: takes blood away from heart or has thick elastic muscular wall (1).
- Vein: carries blood towards heart or has valves (1).
- Capillary: one-cell-thick wall for diffusion/exchange (1).

b) Describe the route of blood from the right atrium, through the lungs, to the left ventricle. **[3]**

- Right atrium to right ventricle (1).
- Via pulmonary artery to lungs, then via pulmonary vein to heart (1).
- Into left atrium, then left ventricle (1).

c) Explain why the left ventricle has a thicker muscular wall than the right ventricle. **[2]**

- The left ventricle pumps blood around the whole body (1).
- It must generate higher pressure than the right ventricle, which pumps only to the lungs (1).

Question 2 Diabetes — 6 marks

a) State where insulin is produced and explain its role in controlling blood glucose. **[2]**

- The pancreas produces insulin (1).
- Insulin helps regulate/reduce blood glucose by promoting glucose uptake/storage (1).

b) Compare Type 1 and Type 2 diabetes. Include the body's response to insulin and ONE fact about how each condition develops. **[4]**

- Type 1: immune system attacks insulin-producing cells (1), so little/no insulin is made (1).
- Type 2: cells respond poorly to insulin/insulin resistance (1); development involves multiple factors such as genetics, body fat distribution and activity, and sometimes reduced insulin production (1).
- Do not award claim that simply eating sugar causes Type 1 or always causes Type 2.

Question 3 Interpreting heart-rate data — 6 marks

Students measure heart rate at several points after activity:

Before activity: 68 beats/min; immediately after: 132 beats/min; 2 minutes later: 96 beats/min; 4 minutes later: 74 beats/min.

a) Calculate the increase from before activity to immediately after activity. Give units. **[1]**

- 132 - 68 = 64 beats/min (1).

b) Describe how heart rate changes over the four minutes after activity. **[2]**

- Heart rate decreases over time (1).
- Uses data for comparison: 132 immediately after to 74 at four minutes (1).

c) Explain why heart rate increases during activity using glucose, oxygen, blood and aerobic respiration. **[3]**

- Working muscles need to release more energy (1).
- Aerobic respiration requires glucose and oxygen (1).
- A higher heart rate increases blood delivery of these reactants and removal of carbon dioxide (1).

Question 4 A blocked coronary artery — 4 marks

a) State the role of the coronary arteries. **[1]**

- They supply oxygenated blood to heart muscle (1).

b) Explain how a coronary artery blockage can cause heart muscle cells to die. **[3]**

- Less blood/oxygen reaches heart muscle cells (1).
- Less aerobic respiration and energy release occurs (1).
- Cells cannot maintain their life processes and may be damaged or die (1).

Question 5 Extended response: linking the systems — 6 marks

a) A student eats a carbohydrate-rich meal, then walks briskly to school. Explain how the digestive, gas-exchange and circulatory systems supply a leg muscle cell with the substances it needs to release energy. Use a logical sequence and scientific words. **[6]**

- Carbohydrates are digested to glucose by digestive processes (1).
- Glucose is absorbed into blood through the small intestine/villi (1).
- Oxygen diffuses from alveoli into the blood (1).
- The heart pumps blood carrying glucose and oxygen to muscle capillaries/cells (1).

- Cells use glucose and oxygen in aerobic respiration (1).
- Energy is released for muscle contraction, and carbon dioxide/water are produced (1).

Total: 30 marks. Mark each valid point once unless stated otherwise.